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1 // Test Case: Impulse Response of a Continuous-Time System
2 // Define time range and sampling interval
3 t_min = -1; // Start time
4 t_max = 5; // End time
5 dt = 0.01; // Time step
6 t = t_min:dt:t_max; // Time vector
7 // Define the impulse function
8 x_impulse = zeros(1, length(t));
9 x_impulse(t == 0) = 1; // Approximate impulse function
10 // Initialize the output signal with zeros
11 y_impulse = zeros(1, length(t));
12 // Define the system parameters
13 a = 2; b = 1; // Coefficient for y(t)
14 // Coefficient for x(t)
15 // Compute the impulse response using the Laplace Transform approach
16 // Analytical solution: y(t) = (b/a) * exp(-a*t) for t >= 0
17 for i = 1:length(t)
18 if t(i) >= 0
19 y_impulse(i) = (b / a) * exp(-a * t(i));
20 end
21 end
22 // Plot the input and output signals for impulse response
23 clf; // Clear the current figure
24 subplot(2,1,1);
25 plot(t, x_impulse, 'b');
26 xlabel('t (Time)');
27 ylabel('x(t)');
28 title('Input Signal x(t) - Impulse Function');
29 subplot(2,1,2);
30 plot(t, y_impulse, 'r');
31 xlabel('t (Time)');
32 ylabel('y(t)');
33 title('Impulse Response y(t) of the System');
34
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